

Comprehensive Validation Analysis

Thesis-Ready Interpretation of the Additive Bayesian Modelling Pipeline

Bayesian Football Analytics Pipeline

Generated 2026-05-14

Abstract

This note summarizes the validation state of the real diagonal, simulated diagonal, and simulated low-rank recovery models. The main thesis result is a descriptive variance decomposition of multivariate football player-season outcomes into player, season, and residual components. The real-data diagonal additive model is usable with mild convergence caveats. Simulation results support the diagonal variance-decomposition machinery for player and residual components and, more cautiously, for season components. The low-rank recovery experiment is an important negative result: it samples without Hamiltonian Monte Carlo geometry failures but does not identify loading matrices reliably. PSIS-LOO was computed, but Pareto- k diagnostics make the approximation unreliable for formal model selection.

1 Motivation and Research Question

The scientific aim is not simply to fit a model that runs. The aim is to establish which football-performance quantities can be interpreted as stable player-level variation, which can be interpreted as season-level variation, and which remain mostly unexplained row-level variation after accounting for players and seasons.

The central research question is:

For a multivariate panel of standardized football features, how much variation is attributable to stable player differences, common season shifts, and player-season residual variation?

The analysis conditions on the observed Wyscout-derived player-season panel. It does not model transfers, injuries, tactical systems, position, team identity, coach identity, match-level opponent strength, or minutes selection beyond the existing filtering. Therefore a player effect should be interpreted as a stable player-season profile effect within the observed data-generating and selection process, not as a pure causal ability parameter.

2 Data Structure and Estimands

2.1 Panel Structure

The modelling unit is an observed player-season row. Missing player-seasons are not imputed. This choice avoids assuming that absence from the panel is ignorable. In football data, absence may be

caused by transfers, injuries, relegation, age, retirement, or selection. The cost is that conclusions apply to observed eligible player-seasons, not to all possible player-season combinations.

Table 1: Observed model object summary.

Quantity	Value
Observed player-season rows	4,944
Selected features	53
Players	1,650
Seasons	12
One-season players	605
Repeated players	1,045
Players with gaps	346
Feature matrix rank	53
Maximum absolute feature correlation	0.972

The feature matrix is full rank. The maximum retained absolute feature correlation is high, mainly reflecting closely related pass-volume variables. This does not break the diagonal additive model, but it warns that some features carry nearly redundant information.

2.2 Notation

Table 2: Core notation used in the validation analysis.

Symbol	Meaning
$n = 1, \dots, N$	Observed player-season row index.
$p = 1, \dots, P$	Feature index.
$i = 1, \dots, I$	Player index.
$j = 1, \dots, S$	Season index.
N	Number of observed player-season rows; here $N = 4,944$.
P	Number of selected scaled football features; here $P = 53$.
I	Number of players; here $I = 1,650$.
S	Number of seasons; here $S = 12$.
$i[n]$	Player associated with row n .
$j[n]$	Season associated with row n .
$\mathbf{y}_n \in \mathbb{R}^P$	Full multivariate response for row n .
y_{np}	Value of feature p in row n .
$\boldsymbol{\mu} \in \mathbb{R}^P$	Population mean in scaled feature space.
$\mathbf{a}_i \in \mathbb{R}^P$	Player effect for player i .
$\mathbf{b}_j \in \mathbb{R}^P$	Season effect for season j .
$\boldsymbol{\epsilon}_n \in \mathbb{R}^P$	Residual player-season effect for row n .
$\boldsymbol{\Sigma}_a$	Player-effect covariance matrix.
$\boldsymbol{\Sigma}_b$	Season-effect covariance matrix.
$\boldsymbol{\Sigma}_\epsilon$	Residual covariance matrix.

Symbol	Meaning
$\mathbf{\Lambda}_a, \mathbf{\Lambda}_b$	Player and season low-rank loading matrices.
$\mathbf{\Psi}_a, \mathbf{\Psi}_b$	Player and season uniqueness variance vectors.

The response vector is

$$\mathbf{y}_n = (y_{n1}, \dots, y_{nP})^\top, \quad (1)$$

where each column has been standardized before modelling. Standardization makes variance components comparable across heterogeneous football features such as shots per 90 minutes, pass rates, recoveries, and card counts.

2.3 Model Components

Table 3: Model components and their interpretation.

Component	Interpretation
μ	Population-level average scaled football profile.
$\mathbf{a}_i$	Stable player-specific deviation from the population profile.
$\mathbf{b}_j$	Common season-specific deviation shared across observed rows.
ϵ_n	Remaining row-level variation after player and season effects.
$\sigma_{a,p}$	Player-effect standard deviation for feature p .
$\sigma_{b,p}$	Season-effect standard deviation for feature p .
$\sigma_{\epsilon,p}$	Residual standard deviation for feature p .
$\text{diag}(\sigma^2)$	Diagonal covariance matrix with one variance per feature.
$\mathbf{\Lambda}\mathbf{\Lambda}^\top$	Low-rank covariance contribution from latent factors.
$\mathbf{\Psi}$	Feature-specific uniqueness variance not explained by common factors.

2.4 Estimands

For each feature p , the real diagonal additive model estimates

$$V_{\text{player},p} = \sigma_{a,p}^2, \quad (2)$$

$$V_{\text{season},p} = \sigma_{b,p}^2, \quad (3)$$

$$V_{\text{residual},p} = \sigma_{\epsilon,p}^2. \quad (4)$$

The proportional variance decomposition is

$$\pi_{\text{player},p} = \frac{V_{\text{player},p}}{V_{\text{player},p} + V_{\text{season},p} + V_{\text{residual},p}}, \quad (5)$$

$$\pi_{\text{season},p} = \frac{V_{\text{season},p}}{V_{\text{player},p} + V_{\text{season},p} + V_{\text{residual},p}}, \quad (6)$$

$$\pi_{\text{residual},p} = \frac{V_{\text{residual},p}}{V_{\text{player},p} + V_{\text{season},p} + V_{\text{residual},p}}. \quad (7)$$

These proportions are the most interpretable thesis outputs. They answer how much of each feature's variation is stable across players, how much is shared by seasons, and how much remains row-specific after accounting for player and season. They are model-based variance components, not causal effects.

3 Real Diagonal Additive Model

The primary real-data model is

$$\mathbf{y}_n = \boldsymbol{\mu} + \mathbf{a}_{i[n]} + \mathbf{b}_{j[n]} + \boldsymbol{\epsilon}_n. \quad (8)$$

The diagonal covariance assumptions are

$$\mathbf{a}_i \sim \mathcal{N}(\mathbf{0}, \text{diag}(\boldsymbol{\sigma}_a^2)), \quad (9)$$

$$\mathbf{b}_j \sim \mathcal{N}(\mathbf{0}, \text{diag}(\boldsymbol{\sigma}_b^2)), \quad (10)$$

$$\boldsymbol{\epsilon}_n \sim \mathcal{N}(\mathbf{0}, \text{diag}(\boldsymbol{\sigma}_\epsilon^2)). \quad (11)$$

The player and season effects are non-centered and column-centered in the pipeline so that $\boldsymbol{\mu}$ remains identifiable. The model is a deliberate first-order variance decomposition. Its strength is interpretability: each feature receives a player, season, and residual variance share. It is not intended to answer the separate question of latent cross-feature covariance structure.

4 Simulation Design

The simulation uses the same row count, feature count, player mapping, and season mapping as the observed panel.

The simulated data are generated from known low-rank plus diagonal player and season covariance components. This creates two validation exercises. First, the diagonal additive model is fitted to knowingly misspecified low-rank data to test whether diagonal variance components are still recovered. Second, a low-rank model is fitted to evaluate whether the known loadings, uniquenesses, and covariance diagonals can be recovered.

Table 4: Simulation design summary.

Quantity	Value
Simulation seed	20260913
Rows	4,944
Features	53
Players	1,650
Seasons	12
Player low-rank dimension	2
Season low-rank dimension	1

5 Low-Rank Recovery Model

The low-rank recovery model uses

$$\Sigma_a = \Lambda_a \Lambda_a^\top + \text{diag}(\Psi_a), \quad (12)$$

$$\Sigma_b = \Lambda_b \Lambda_b^\top + \text{diag}(\Psi_b), \quad (13)$$

$$\Sigma_\epsilon = \text{diag}(\sigma_\epsilon^2). \quad (14)$$

The model estimates player loadings Λ_a , season loadings Λ_b , player uniqueness scales, season uniqueness scales, residual scales, and the implied covariance diagonals. It uses pure-anchor constraints: selected anchor variables have positive own-factor loadings and zero off-factor loadings. These constraints reduce rotation and sign ambiguity, but the results show that they do not fully solve practical identifiability.

6 Diagnostics and Posterior Health

6.1 Diagnostic Definitions

Table 5: Diagnostics and validation metrics.

Diagnostic	Interpretation
$\hat{R}$	Chain convergence diagnostic. Values close to 1 are good; values above 1.01 indicate possible mixing issues.
Bulk ESS	Effective sample size for the central part of the posterior.
Tail ESS	Effective sample size for posterior tails.
Divergence	HMC warning that trajectories had difficulty following posterior geometry.
Treedepth hit	NUTS reached the configured maximum tree depth.
E-BFMI	Energy Bayesian Fraction of Missing Information; low values indicate poor energy exploration.
PSIS-LOO	Pareto-smoothed importance-sampling approximation to leave-one-out cross-validation.

Diagnostic	Interpretation
Pareto- k	Stability diagnostic for PSIS weights; $k > 0.7$ is problematic and $k > 1$ is severe.
MAE	Mean absolute error between posterior estimate and simulation truth.
RMSE	Root mean squared error between posterior estimate and simulation truth.
Coverage	Proportion of true values inside the central 90% posterior interval.

Posterior diagnostics are part of the scientific argument. Parameters with clean $\hat{R}$ and adequate ESS can be interpreted normally. Parameters with mild warnings should be interpreted cautiously. Parameters with severe warnings should not be used for substantive conclusions.

6.2 Sampler Geometry

Table 6: Sampler geometry summary for the three fitted models.

Model	Draws	Div.	Max TD	TD hits	Min E-BFMI	Mean accept
Real diagonal	4,000	0	8	0	0.561	0.953
Simulated diagonal	4,000	0	8	0	0.551	0.951
Simulated low-rank	4,000	0	8	0	0.586	0.951

All three models have clean HMC geometry: no divergences, no treedepth saturations at the configured limit of 12, E-BFMI above common warning thresholds, and average acceptance rates close to the target `adapt_delta` = 0.95. The main problems are therefore not HMC geometry problems. They are posterior mixing, identifiability, and predictive validation diagnostics.

6.3 Posterior Health

Table 7: Posterior summary diagnostics.

Model	Parameters	Max $\hat{R}$	> 1.01	> 1.05	Min bulk ESS	Min tail ESS
Real diagonal	531	1.035	13	0	159	175
Simulated diagonal	531	1.010	3	0	779	1,412
Simulated low-rank	688	1.739	233	208	6	107

The real diagonal model is mostly usable, with mild mixing warnings in a small number of player variance parameters. The simulated diagonal model is healthy. The simulated low-rank model is not healthy for loading interpretation.

No real-data $\hat{R}$ exceeds 1.05. For thesis reporting these parameters should be described as mildly under-mixed rather than perfectly converged.

Table 8: Worst real diagonal posterior diagnostics.

Parameter	Feature	Mean	SD	$\hat{R}$	Bulk ESS	Tail ESS
var_a[11]	per90_forward_passes	0.827	0.032	1.035	159	290
sigma_a[11]	per90_forward_passes	0.909	0.018	1.035	159	290
prop_e[11]	per90_forward_passes	0.150	0.006	1.028	202	642
prop_a[11]	per90_forward_passes	0.846	0.006	1.027	179	376
var_a[44]	per90_sliding_tackles	0.652	0.029	1.016	894	1,643
var_a[16]	per90_progressive_passes	0.752	0.029	1.015	509	1,746
var_a[24]	rate_successful_crosses	0.022	0.013	1.012	197	177

Table 9: Worst low-rank posterior diagnostics.

Parameter	Mean	SD	$\hat{R}$	Bulk ESS
lambda_b_free[37] / $\Lambda_b[38, 1]$	-0.010	0.271	1.739	6.1
lambda_a_free[41] / $\Lambda_a[43, 1]$	0.000	0.111	1.738	6.2
lambda_a_free[23] / $\Lambda_a[25, 1]$	0.000	0.210	1.738	6.1
lambda_b_free[24] / $\Lambda_b[25, 1]$	0.018	0.328	1.738	6.1
psi_b[1]	0.332	0.246	1.737	6.2
psi_a[1]	0.418	0.127	1.737	6.2

These values are a strong identifiability and mixing warning. Chains are not exploring the same posterior region for many loading and uniqueness parameters, even though HMC trajectories themselves are clean.

7 Real-Data Variance Decomposition

The real diagonal additive model decomposes scaled feature variance into player, season, and residual components. Averaged across features, the player component is dominant.

Table 10: Mean variance shares across the 53 real-data features.

Component	Mean share
Player	0.634
Season	0.047
Residual	0.319

This means that stable player-level heterogeneity is the dominant source of variation in most selected features. Season-level variation exists but is much smaller on average. Residual variation remains important, especially for rate features.

7.1 Features Dominated by Player Effects

Volume and style variables are strongly player-specific. This is consistent with stable player roles, positions, and tactical usage.

Table 11: Highest player-level variance shares.

Feature	Player	Season	Residual
per90_recoveries	0.884	0.016	0.100
per90_touch_in_box	0.872	0.001	0.127
per90_offensive_duels	0.869	0.056	0.075
per90_shots	0.860	0.008	0.132
per90_forward_passes	0.846	0.004	0.150
per90_crosses	0.846	0.006	0.148
per90_long_passes	0.843	0.005	0.152
per90_clearances	0.837	0.034	0.129
per90_passes	0.829	0.011	0.160
per90_back_passes	0.823	0.008	0.169

7.2 Features With the Largest Season Effects

Table 12: Highest season-level variance shares.

Feature	Player	Season	Residual
per90_smart_passes	0.440	0.376	0.183
per90_pressing_duels	0.455	0.320	0.225
per90_accelerations	0.598	0.164	0.238
per90_dribbles_against	0.693	0.136	0.171
rate_defensive_duels_won	0.244	0.129	0.627
per90_through_passes	0.583	0.121	0.296
per90_sliding_tackles	0.606	0.119	0.275
per90_loose_ball_duels	0.718	0.113	0.168
per90_dribbles	0.800	0.101	0.100
rate_successful_progressive_passes	0.484	0.084	0.432

The strongest season effects occur in features that plausibly reflect changing tactical environments, league-wide style, event definitions, or tactical fashion over seasons. The most prominent examples are smart passes and pressing duels.

7.3 Features Dominated by Residual Effects

Rate variables and sparse outcome variables are noisy. They depend strongly on opportunity counts and row-level context and are much less stable than volume/style features.

8 Simulation Recovery Results

8.1 Diagonal Model on Low-Rank Data

The simulated diagonal model is intentionally misspecified because the data were generated from low-rank plus diagonal covariance. Despite this, it recovers diagonal variance components well.

Table 13: Highest residual variance shares.

Feature	Player	Season	Residual
rate_successful_crosses	0.022	0.003	0.975
rate_successful_sliding_tackles	0.057	0.011	0.932
rate_shots_on_target	0.167	0.001	0.832
rate_goal_conversion	0.180	0.003	0.818
rate_dribbles_against_won	0.184	0.008	0.808
rate_successful_dribbles	0.231	0.009	0.760
rate_defensive_duels_won	0.244	0.129	0.627
per90_assists	0.391	0.010	0.599
rate_offensive_duels_won	0.412	0.016	0.573
per90_yellow_cards	0.441	0.017	0.542

Table 14: Recovery metrics for the simulated diagonal additive model.

Component	Bias	MAE	RMSE	Corr.	90% coverage
Player	0.0028	0.0078	0.0112	0.996	0.906
Residual	-0.0009	0.0040	0.0055	0.998	0.925
Season	0.0250	0.0300	0.0468	0.885	0.868

Player variance recovery is excellent, residual variance recovery is excellent, and season variance recovery is weaker but still useful. The season component is hardest because there are only 12 seasons, which is a small number of season-level units for feature-by-feature season variance estimation.

Table 15: Worst season variance errors in the simulated diagonal model.

Feature	Posterior mean	Truth	Error
rate_successful_sliding_tackles	0.354	0.139	0.214
per90_xg_shot	0.221	0.103	0.118
per90_goals	0.236	0.151	0.085
per90_fouls	0.144	0.064	0.080
per90_defensive_duels	0.130	0.057	0.073

The bias is mostly upward for season variance, which is consistent with a diagonal model approximating low-rank seasonal covariance using feature-specific season variance terms.

8.2 Low-Rank Loading Recovery

The player loading matrix has moderate correlation with truth but large errors and poor mixing. The season loading matrix is not recovered well: its correlation with truth is only 0.262, and coverage is high because intervals are too wide, not because estimates are precise.

Many free loadings have posterior means near zero, wide intervals, and very high $\hat{R}$. The current low-rank model should therefore be treated as a failed or incomplete loading-recovery experiment, not as a substantive football-factor model.

Table 16: Low-rank loading recovery metrics.

Parameter	MAE	RMSE	Corr.	90% coverage
Λ_a	0.082	0.131	0.804	0.840
Λ_b	0.102	0.124	0.262	1.000

Table 17: Worst Λ_a loading errors.

Feature	Factor	Mean	Truth	Error	$\hat{R}$
per90_shot_assists	1	-0.003	0.491	-0.494	1.736
per90_fouls	1	-0.001	0.352	-0.352	1.736
per90_shots_blocked	1	-0.001	0.340	-0.342	1.735
per90_opponent_half_recoveries	1	0.002	-0.335	0.337	1.734
per90_long_passes	1	0.002	-0.289	0.291	1.735

8.3 Low-Rank Variance and Scale Recovery

The low-rank model recovers residual scales very well, player covariance diagonals very well, and player uniqueness scales reasonably well. It does not recover season covariance diagonals precisely, season uniqueness scales as well as player scales, or loading matrices reliably.

Recovering a covariance diagonal is easier than recovering a loading matrix. Different loading configurations can imply similar total feature variance, so good recovery of $\text{diag}(\Sigma_a)$ does not imply good recovery of Λ_a .

9 PSIS-LOO Interpretation

PSIS-LOO was computed with `BFA_RUN_LOO=true`. The nominal estimates are shown in Table 21. The simulated low-rank model has better nominal predictive fit than the simulated diagonal model, with $\Delta\text{elpd}_{\text{low-rank-diagonal}} \approx 5,576$. This direction is expected because the simulated data were generated from a low-rank process.

The Pareto- k diagnostics are unacceptable: thousands of rows have $k > 0.7$, thousands have $k > 1$, and maximum k is far above 1 for all models. The LOO estimates should therefore not be used as formal model-selection evidence. They are better interpreted as evidence that many rows are highly influential under conditional leave-one-row-out scoring.

Table 18: Worst Λ_b loading errors.

Feature	Factor	Mean	Truth	Error	$\hat{R}$
rate_successful_sliding_tackles	1	-0.042	0.298	-0.340	1.736
per90_xg_shot	1	-0.008	0.230	-0.238	1.737
per90_passes	1	-0.003	0.230	-0.233	1.733
per90_dangerous_own_half_losses	1	-0.008	0.206	-0.214	1.735
per90_through_passes	1	0.023	-0.185	0.208	1.734

Table 19: Low-rank variance and scale recovery metrics.

Parameter	MAE	RMSE	Corr.	90% coverage
$\text{diag}(\Sigma_a)$	0.009	0.013	0.997	0.811
$\text{diag}(\Sigma_b)$	0.057	0.093	0.886	0.585
ψ_a SD	0.011	0.022	0.941	0.925
ψ_b SD	0.040	0.052	0.740	0.925
σ_ϵ SD	0.004	0.005	0.998	0.925

Table 20: Worst $\text{diag}(\Sigma_b)$ errors in the low-rank recovery model.

Feature	Mean	Truth	Error	$\hat{R}$
rate_successful_sliding_tackles	0.572	0.139	0.432	1.051
per90_goals	0.435	0.151	0.284	1.029
per90_xg_shot	0.342	0.103	0.238	1.009
per90_through_passes	0.219	0.084	0.134	1.013
per90_fouls	0.160	0.064	0.096	1.003

Table 21: PSIS-LOO results and Pareto- k diagnostics.

Model	elpd_{loo}	SE	p_{loo}	looic	$k > 0.7$	$k > 1$	Max k
Real diagonal	-224,673.4	1,007.0	49,545.7	449,346.9	4,354	3,088	3.82
Simulated diagonal	-201,763.1	394.7	41,871.8	403,526.1	4,529	2,878	2.88
Simulated low-rank	-196,186.8	363.3	32,963.7	392,373.7	4,215	2,104	2.08

This is unsurprising for a hierarchical player-season model. Many players appear only once, player effects can be informed heavily by a single row, and leaving out that row can substantially change the effective posterior. Exact K -fold cross-validation is preferable for real predictive validation.

10 Limitations

The main limitations are as follows.

1. The real diagonal model has mild mixing issues. The maximum $\hat{R}$ is 1.035, 13 summarized parameters exceed 1.01, the minimum bulk ESS is 159, and the minimum tail ESS is 175.
2. The low-rank recovery model has severe loading-level non-identification. The maximum $\hat{R}$ is 1.739, many loadings have bulk ESS near 6, and many loading posterior means are near zero despite nonzero truth.
3. Season effects are hard to estimate. There are only 12 seasons, and the simulation confirms weaker recovery for season variance and especially season loadings.
4. PSIS-LOO fails Pareto- k diagnostics and should not be used as decisive model-selection evidence.
5. Memory and output size constrain diagnostics. Full fit CSVs are large, and full draw extraction can hit R memory limits.

11 Proposed Fixes and Future Experiments

Several extensions follow directly from the validation evidence.

1. Run a longer real diagonal model with more sampling iterations to check whether the few weak variance parameters improve.
2. Build exact K -fold validation. Leave-player-out folds target generalization to unseen players; leave-season-out folds target generalization to new seasons; random row folds are appropriate only if the target is another observed row from the same player-season process.
3. For simulation only, initialize low-rank parameters near truth to distinguish poor exploration from deeper non-identifiability.
4. Strengthen low-rank anchor constraints by lower-bounding anchor loadings, using stronger anchor priors, selecting higher-signal anchors, and adding stronger shrinkage to free loadings.
5. Simplify the season structure. With only 12 seasons, diagonal season effects or a strongly pooled scalar season score may be more defensible than a feature-level low-rank season loading vector.
6. Add feature sensitivity checks: remove highly correlated pass-volume duplicates, analyze rate-only and volume-only feature sets separately, stratify by position if metadata are available, and test higher minimum-minute thresholds.

7. Add targeted memory-safe diagnostics that extract only a small variable set when trace plots or draw-level checks are needed.

12 Thesis-Ready Conclusions

The pipeline has succeeded for descriptive additive variance decomposition. The real diagonal additive model gives a defensible main result: stable player effects dominate many volume and style features, season effects are generally smaller but meaningful for selected tactical variables, and residual variation dominates sparse rates and conversion or success variables.

The simulation validates the basic additive machinery. The diagonal model recovers player and residual variance components very well, and season variance moderately well, even when the simulated truth contains low-rank covariance.

The current low-rank recovery model should be reported as a diagnostic negative result. It has clean sampler geometry but fails to identify loadings reliably, especially for season effects. Its covariance-diagonal recovery is more credible than its loading recovery, which reinforces the distinction between variance-decomposition claims and latent-factor claims.

PSIS-LOO was computed but fails Pareto- k diagnostics. It should be mentioned only as an attempted predictive diagnostic, not as decisive model-selection evidence. Exact K -fold validation is the appropriate next step for predictive comparison.